



Zeph

User Manual

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Abstract

This document is a user manual for Zeph. It covers installation of the system for local development and provides a step-by-step guide for each user role: passenger, airline, and administrator. It is recommended that readers of this document read the Zeph technical guide document as a prerequisite.

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1. System Requirements & Installation

1.1 Prerequisites

The following must be installed before running Zeph locally:

- Node.js v18 or later
- npm v9 or later
- Git
- MetaMask browser extension
- A Supabase project

1.2 Clone and Install

```
git clone
https://gitlab.com/computing.dcu.ie/sweens47/2026-csc1097-sweens-47-mahons-25.git
```

```
cd 2026-csc1097-sweens-47-mahons-25
```

```
npm install
```

```
npm --prefix src/web install
```

1.3 Environment Configuration

Copy the example environment file and fill in the required values:

```
cp src/web/.env.example src/web/.env.local
```

Open `src/web/.env.local` and set the following variables:

Required for all users:

- **NEXT_PUBLIC_SUPABASE_URL** - your Supabase project URL
- **NEXT_PUBLIC_SUPABASE_ANON_KEY** - your Supabase anon key
- **SUPABASE_URL** - your Supabase project URL (server-side)
- **SUPABASE_SERVICE_ROLE_KEY** - your Supabase service role key
- **NEXT_PUBLIC_CONTRACT_ADDRESS** - the deployed Zeph contract address on Polygon
- **NEXT_PUBLIC_CHAIN_ID** - set to 137 (Polygon mainnet)
- **NEXT_PUBLIC_ADMIN_EMAILS** - comma-separated list of admin email addresses

- **ADMIN_SECRET** - secret key used for admin authentication

Required only if redeploying the smart contract:

- **PRIVATE_KEY** - deployer wallet private key
- **POLYGON_RPC_URL** - RPC endpoint for Polygon

1.4 Running the Project

Start the full application with a single command from the project root:

```
npm run dev
```

This starts two processes simultaneously:

- Next.js UI → the web application, accessible at <http://localhost:3000>
- Oracle worker → a background process that polls for flight delays every 60 seconds and reports eligible claims to the smart contract on Polygon mainnet

Both processes share the environment variables from `src/web/.env.local`, so ensure that file is configured correctly before starting (see Section 1.3).

To stop the application, press **Ctrl+C** - both processes will shut down cleanly

2. Passenger Guide

This section walks through the complete passenger journey. Booking a flight, Registering the flight for compensation, Receiving compensation, Gathering evidence for a rejected claim.

2.1 Booking a Flight

Passengers do not need a Zeph account to book a flight. From the Zeph landing page, click the 'Book a Flight' button.

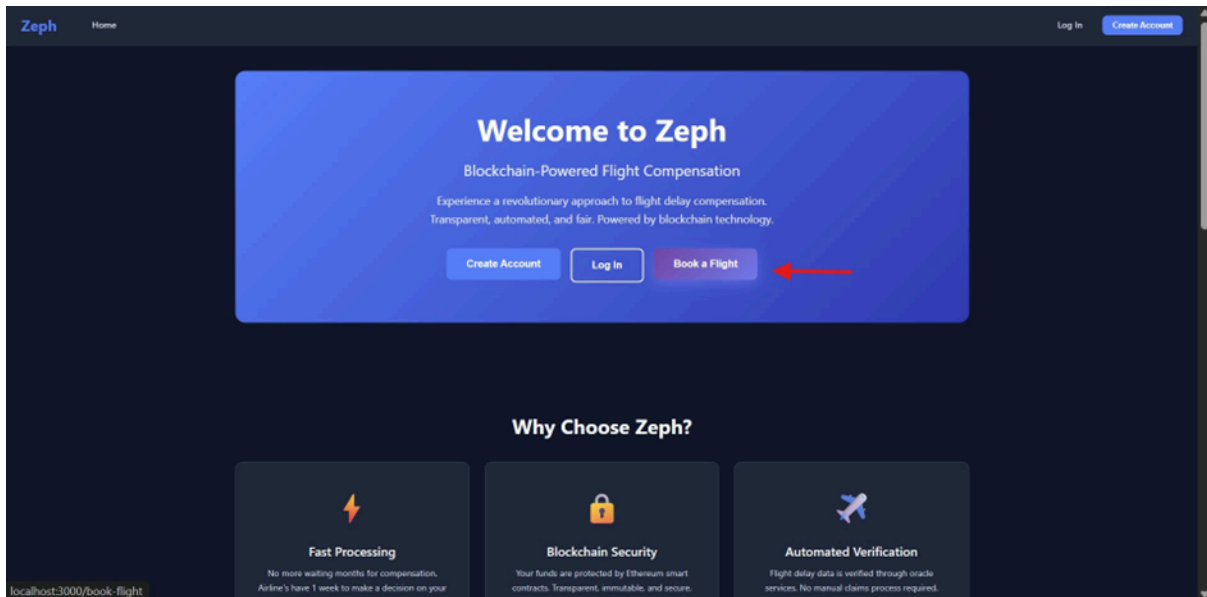


Figure 1 - Zeph landing page. Click 'Book a Flight' to begin.

This represents a user on a third party 'Airline' site where a passenger would book a flight. The booking form opens. Fill in your route, date, departure time, cabin class, airline, passport number, and email address.

Figure 2 - Flight booking form (empty).

Once all fields are filled in, click Complete Booking.

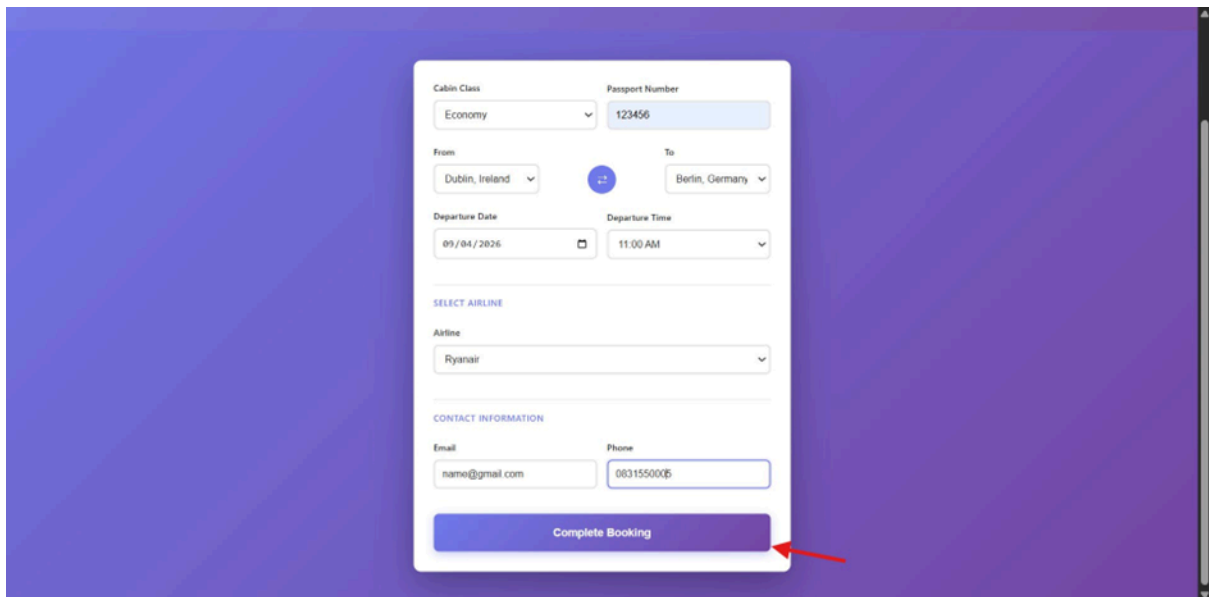


Figure 3 - Booking form completed and ready to submit.

On success, a confirmation modal appears showing your booking reference. Click 'Copy reference' to save it to your clipboard - you will need it when registering your flight for compensation on Zeph.

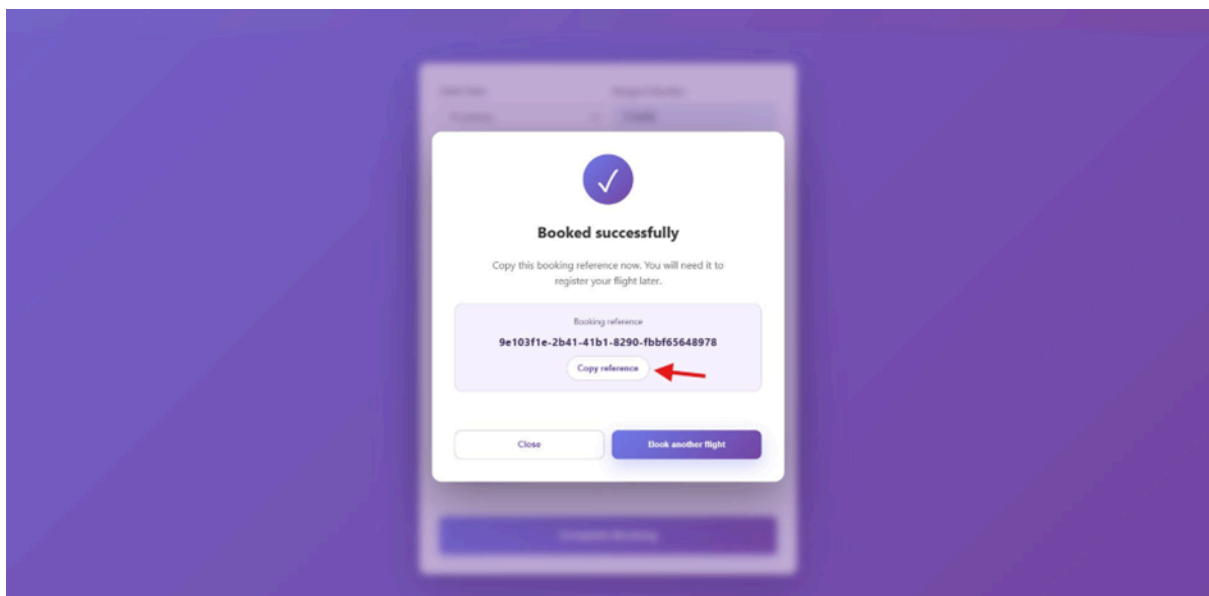


Figure 4 - Booking confirmation modal. Click 'Copy reference'.

The button confirms the reference has been copied. Click 'Close' to dismiss the modal.

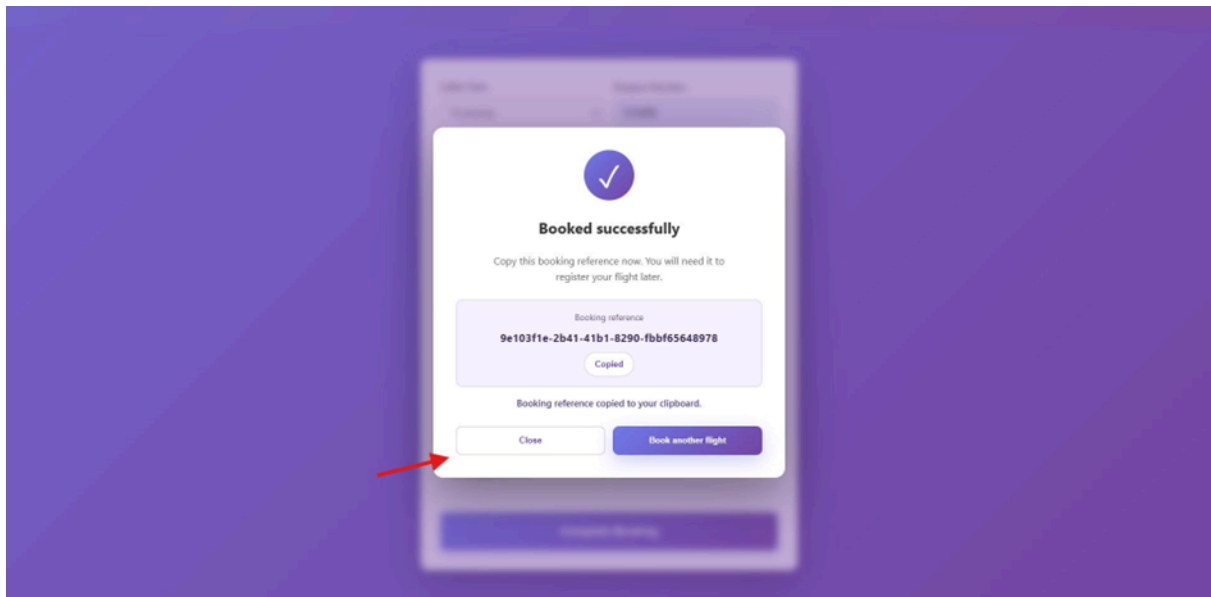


Figure 5 - Reference copied. Click 'Close'.

To register your flight for compensation, return to Zeph and either log in or create an account.

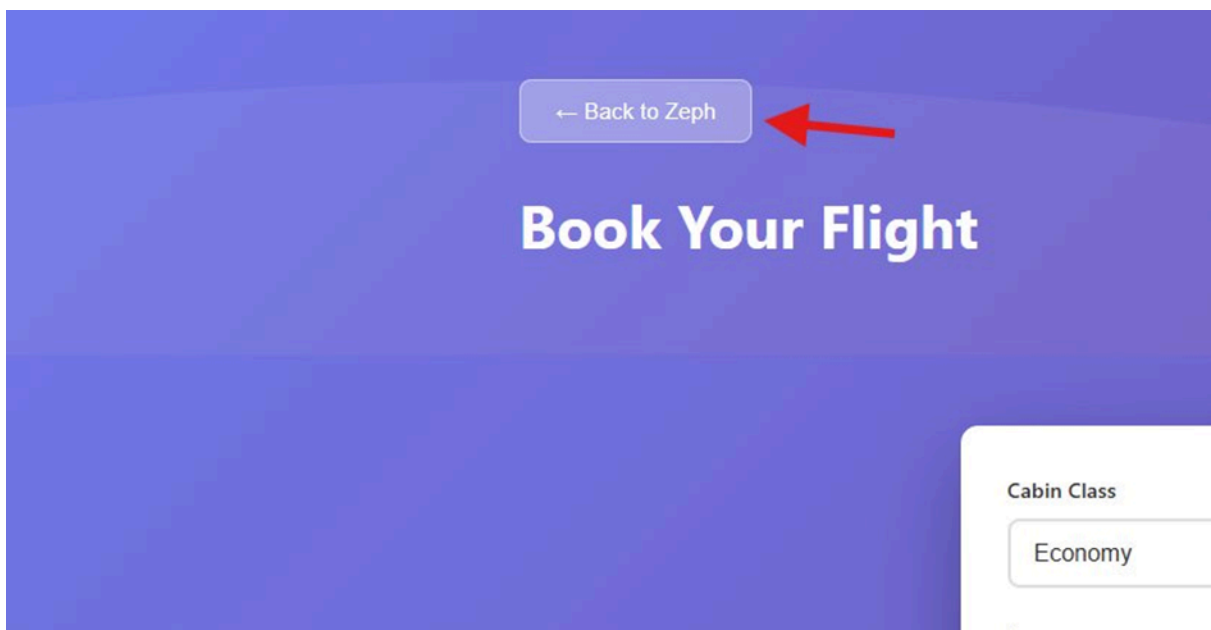


Figure 6 - Return to Zeph after booking.

2.2 Creating an Account

To register a flight for compensation you need a Zeph account. From the landing page, click 'Create Account' in the top right corner.

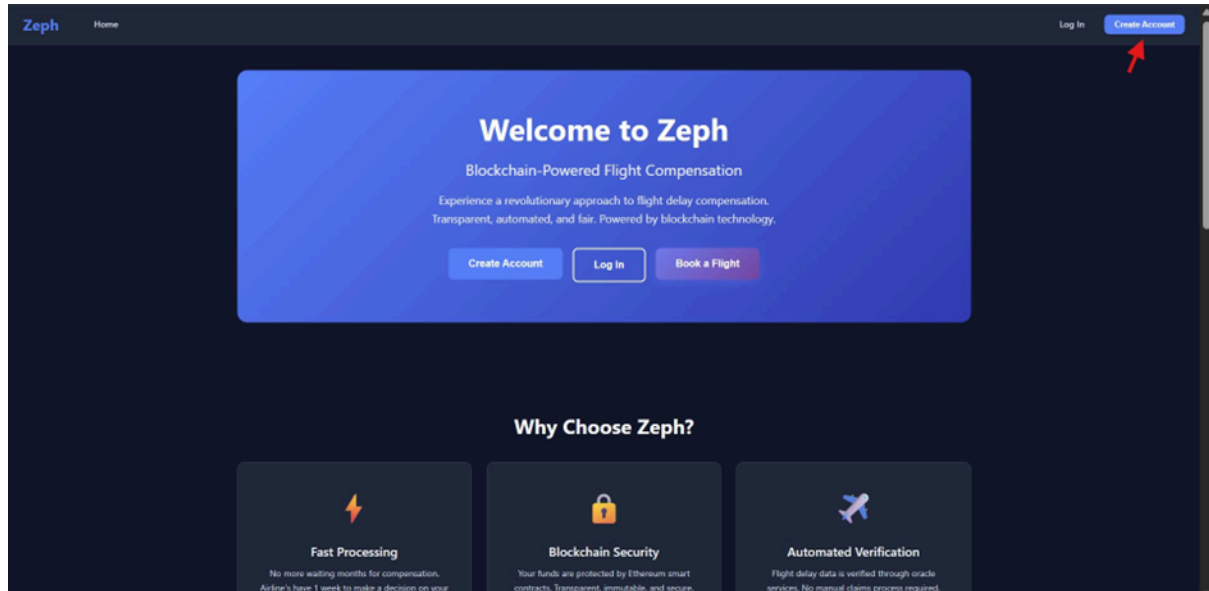


Figure 7 - Landing page. Click 'Create Account'.

Enter your name, email address, and password, then confirm your password and click 'Create Account'.

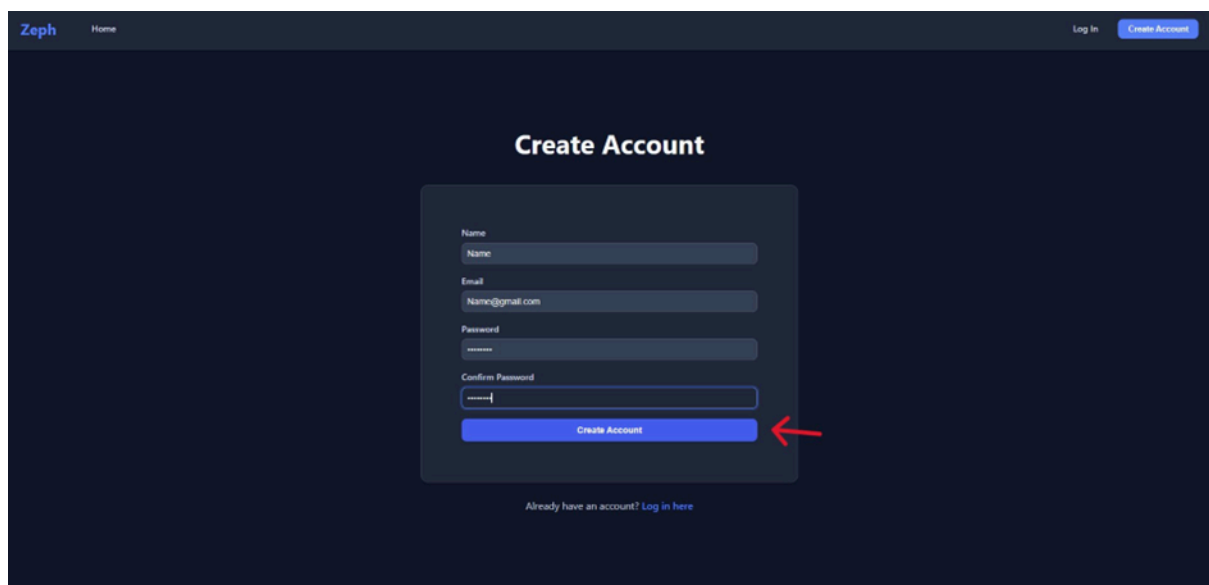


Figure 8 - Account creation form filled in.

2.3 Logging in

After your account is created you will see a success message. Click 'Log in here' to go to the login page.

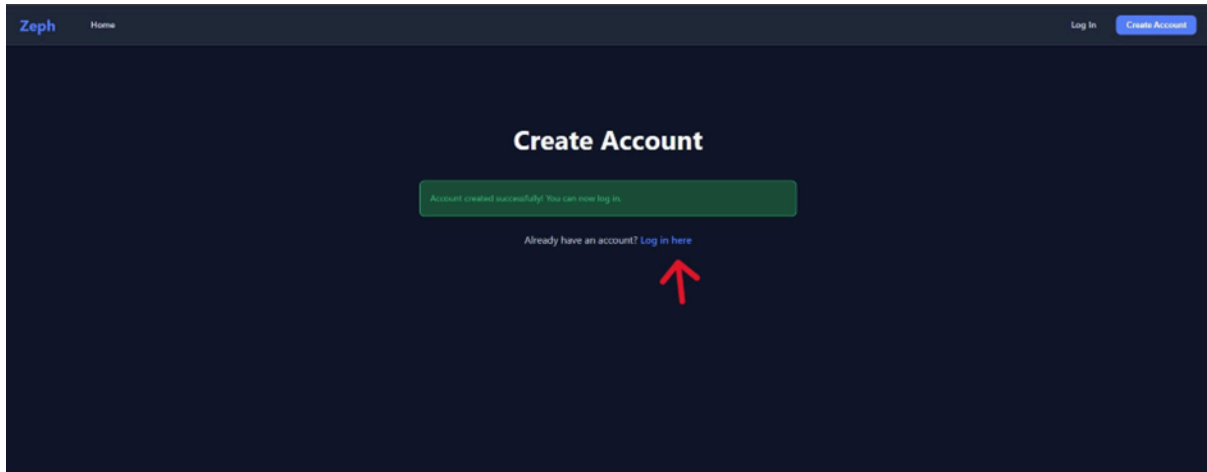


Figure 9 - Account created successfully. Click 'Log in here'.

Enter your email and password, select 'Passenger', and click 'Log In'.

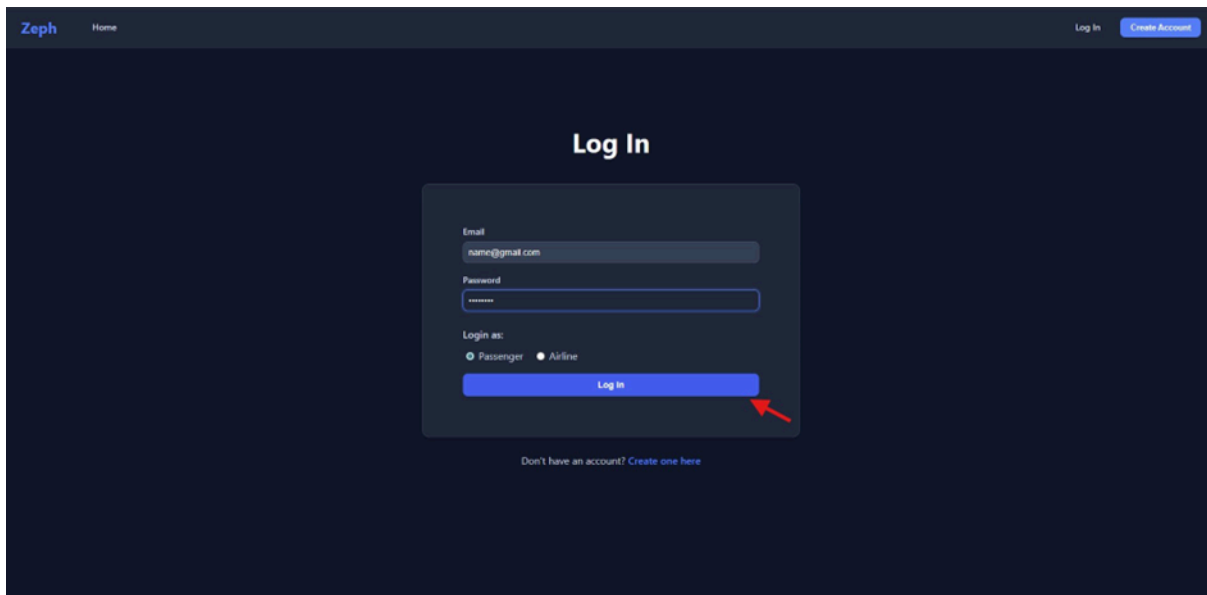


Figure 10 - Login form completed. Click 'Log in'.

2.4 Registering a Flight for Compensation

After logging in you are taken to the Register Flight page.

Paste your booking reference into the 'Booking Reference' field and click 'Register Flight'.

Figure 11 - Register Flight page with booking reference entered.

MetaMask will open and ask you to confirm the blockchain transaction. Review the network and fee details, then click 'Confirm'.

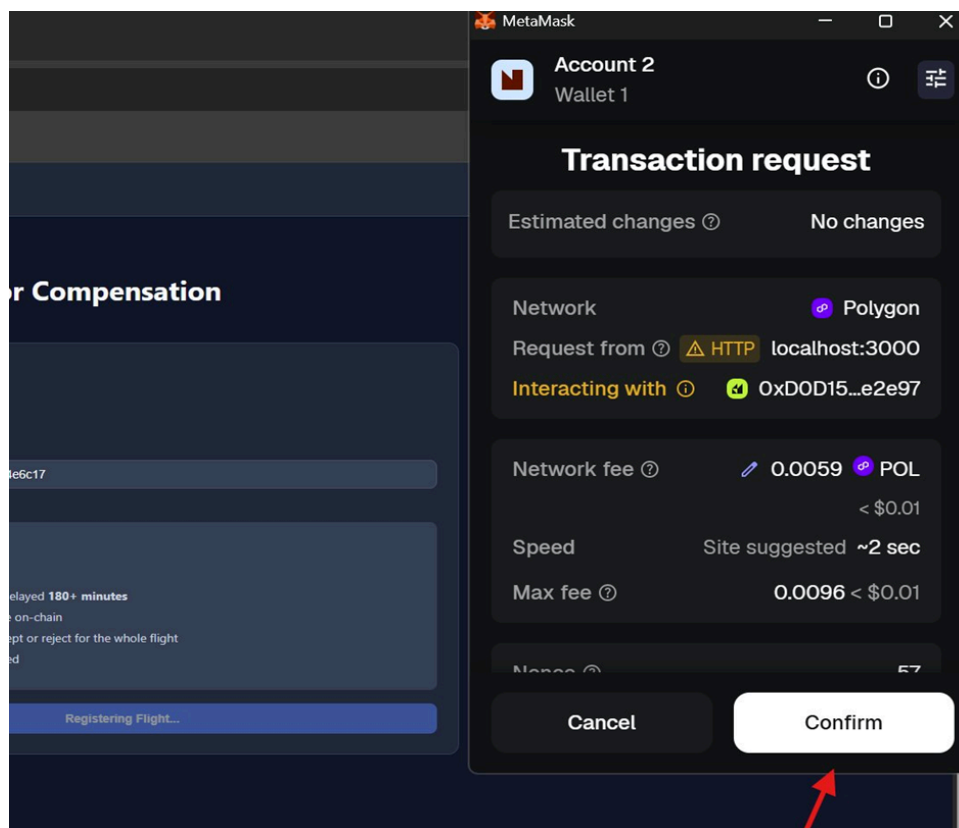


Figure 12 - MetaMask transaction request. Click 'Confirm'.

Once the transaction is confirmed on-chain, the page shows a success banner with the transaction hash. MetaMask also confirms the transaction is complete.

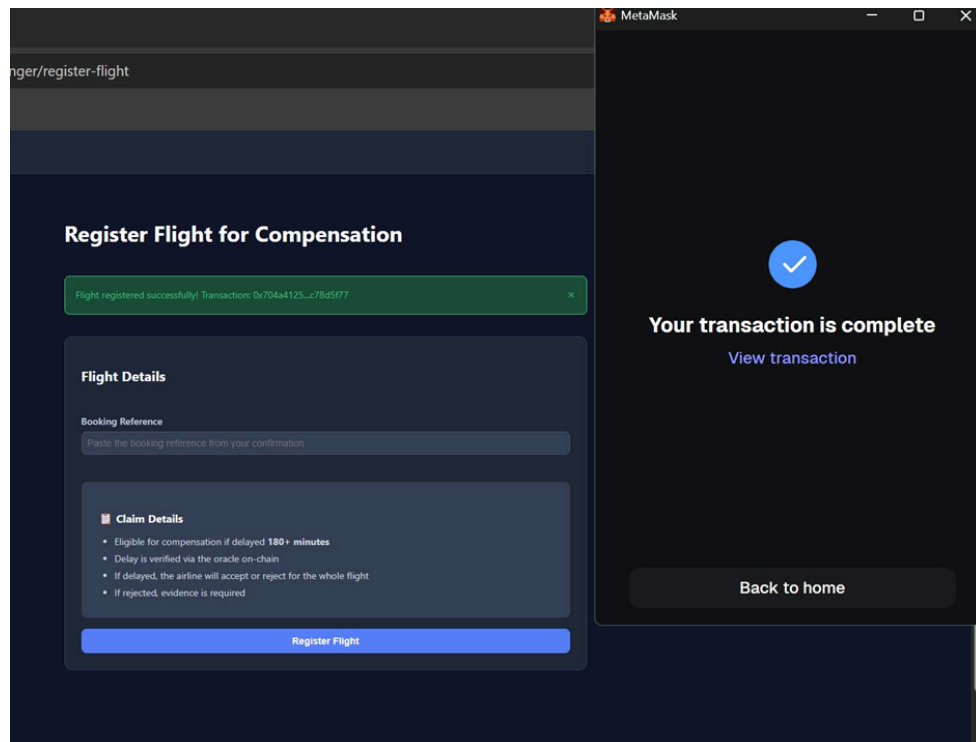


Figure 13 - Transaction confirmed. The flight is now registered on the Polygon blockchain.

2.5 Viewing Claims

Navigate to 'My Claims' using the top navigation bar to see all your registered flights and their current claim status.

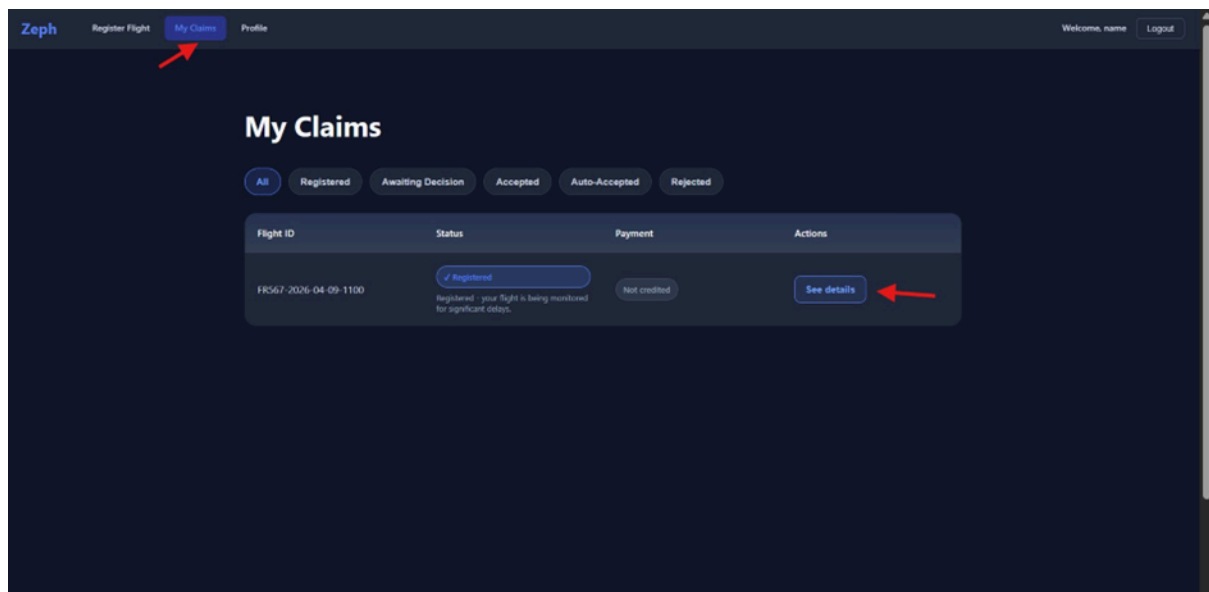


Figure 14 - My Claims page showing the registered flight.

Claims can be filtered by status using the buttons at the top of the page: All, Registered, Awaiting Decision, Accepted, Auto-Accepted, and Rejected. The claim statuses are explained in the appendix

2.6 Claim Details and Blockchain Verification

Click 'See details' on any claim to open the claim details modal. This shows full flight information, claim status, and all associated blockchain transaction hashes.

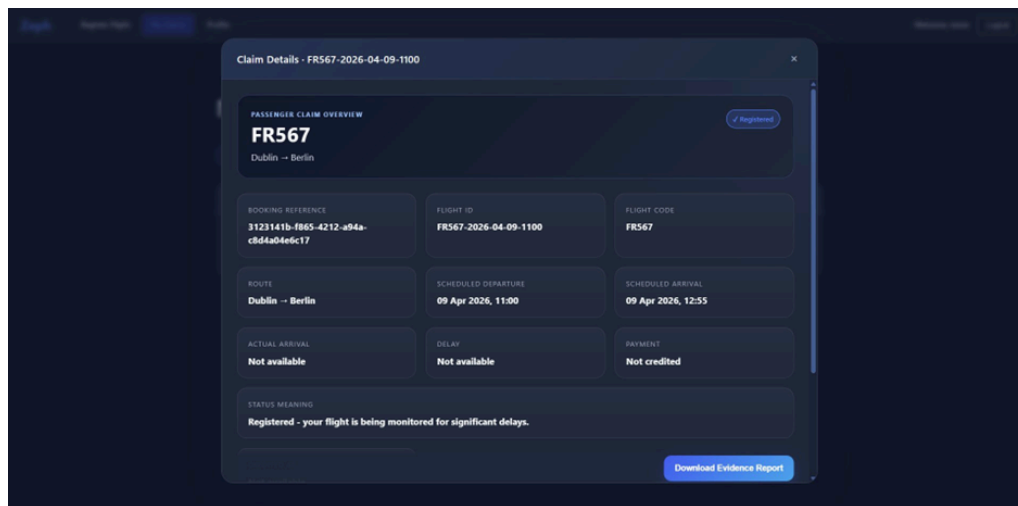


Figure 15 - Claim details modal showing flight data and current status.

Scroll down in the modal to see the Registration Transaction field, which contains the on-chain transaction hash for your claim registration.

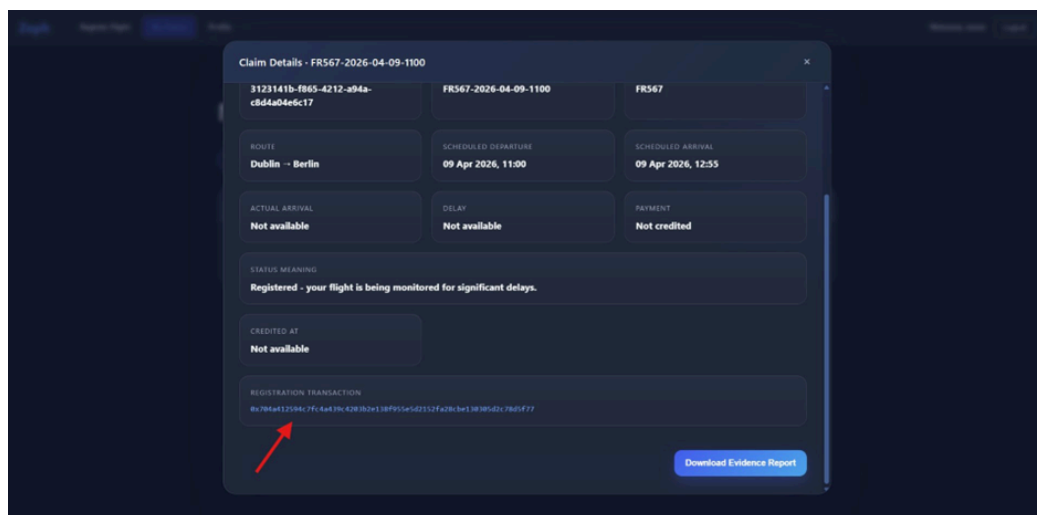


Figure 16 - Claim details modal scrolled down, showing the Registration Transaction hash.

Click the transaction hash to open Polygonscan and independently verify the transaction on the public blockchain.

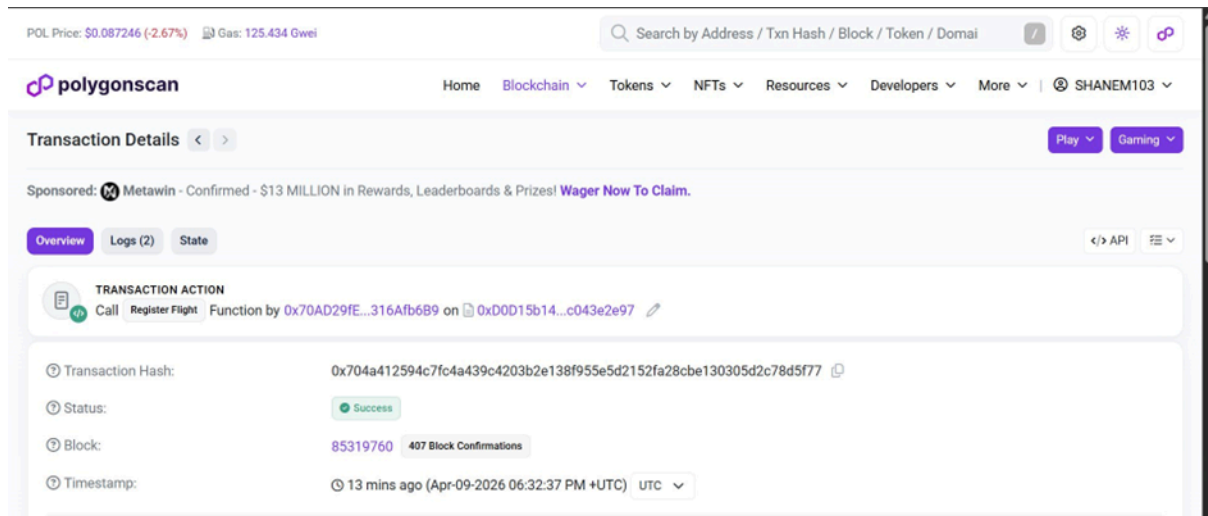


Figure 17 - Polygonscan showing the confirmed registerFlight transaction.

2.7 Claim Rejection Scenario

If your claim is rejected you will see the status of your claim change to 'Rejected'

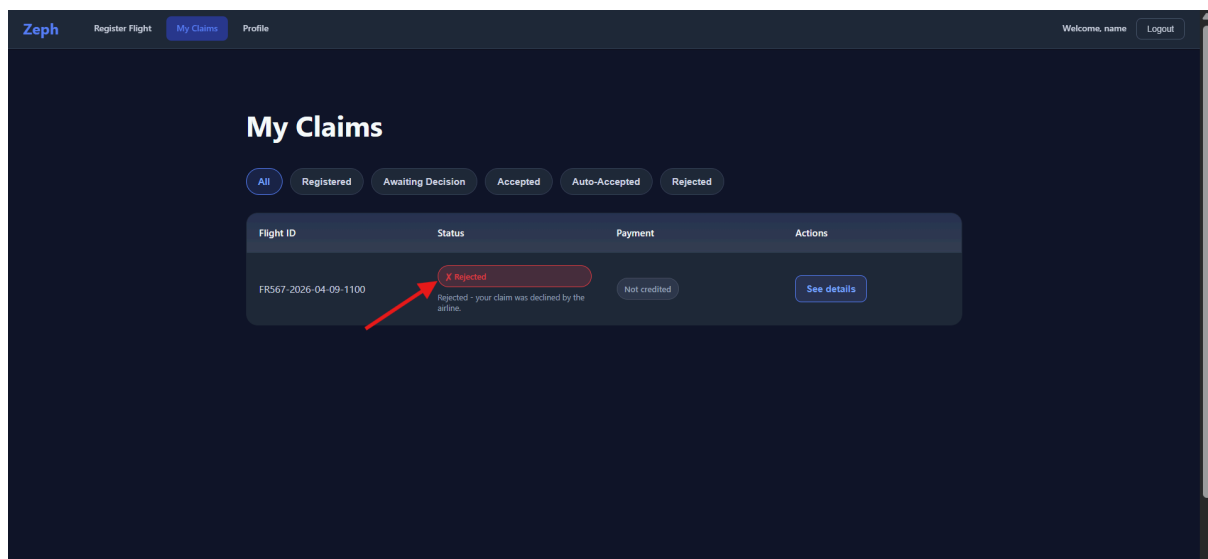


Figure 18 - Claim status updated to 'Rejected'

If you wish to view the details of your claim rejection and/or collect all the relevant information about your claim you can do so. You may decide to escalate your claim to the relevant authorities in your jurisdiction, you can use this evidence report as your evidence. Open the claim details modal as was shown in section 2.6, scroll down and click 'Download Evidence Report'. You can also verify the delay transaction and view the airline's report on why the claim was rejected.

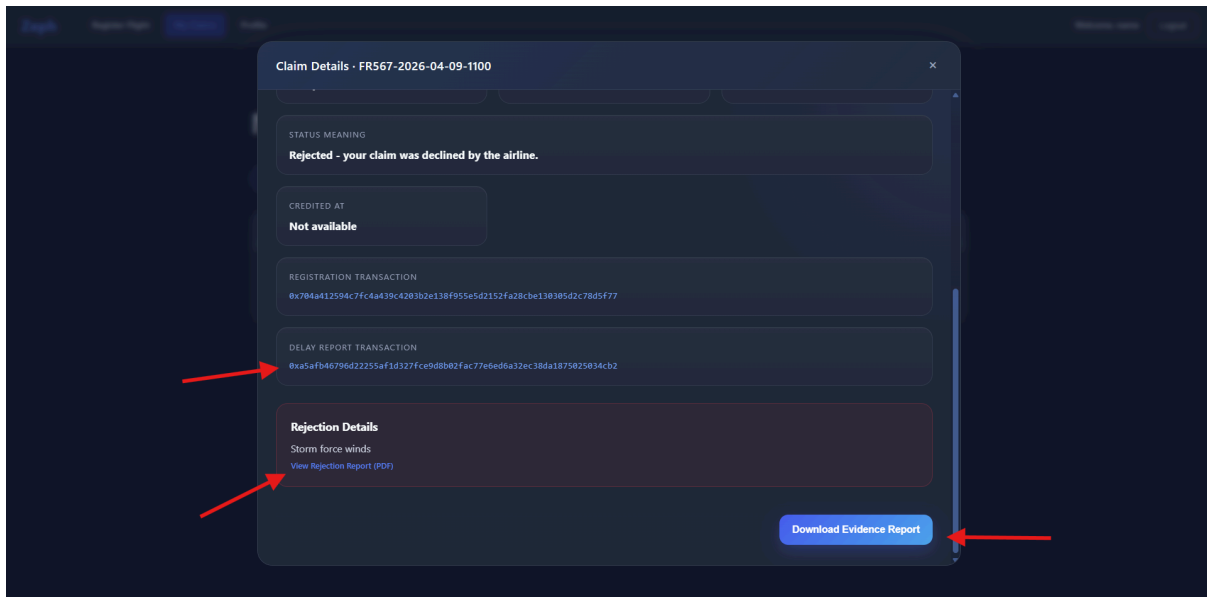


Figure 19 - Click 'Download Evidence Report' to collect your evidence report ZIP.

This evidence report will be downloaded to your device. Click on this to open and view the zip contents.

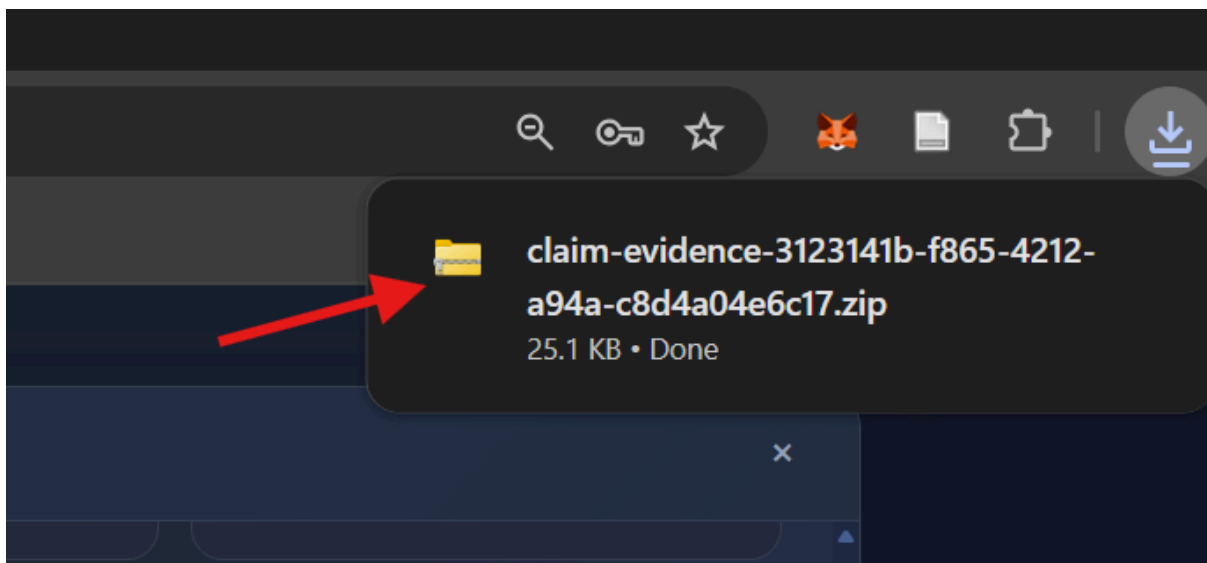
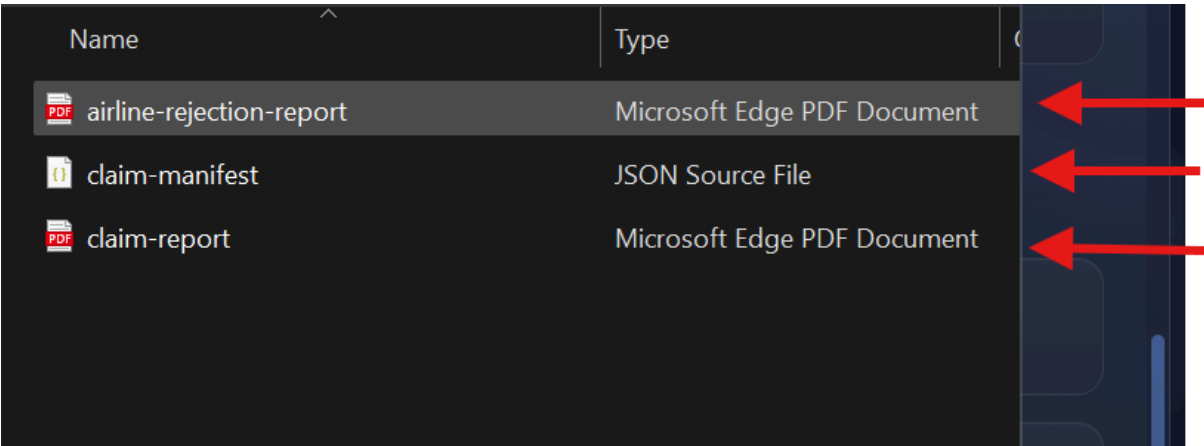


Figure 20 - The evidence report ZIP will be downloaded to your device.

The report ZIP contains three files for a rejection:

1. 'claim-report.pdf' A PDF summary of the claim, including flight details, delay information, claim status, airline decision, and blockchain transaction hashes for independent verification on PolygonScan. Includes a SHA-256 checksum of the evidence payload to prove the data has not been tampered with
2. 'claim-manifest.json' A machine-readable version of the same data in JSON format.

3. 'airline-rejection-report .pdf' (only present if the claim was rejected) The document uploaded by the airline stating their reason for rejection. It's checksum is recorded for verification purposes.



Name	Type
 airline-rejection-report	Microsoft Edge PDF Document
 claim-manifest	JSON Source File
 claim-report	Microsoft Edge PDF Document

Figure 21 - Contents of evidence report ZIP.

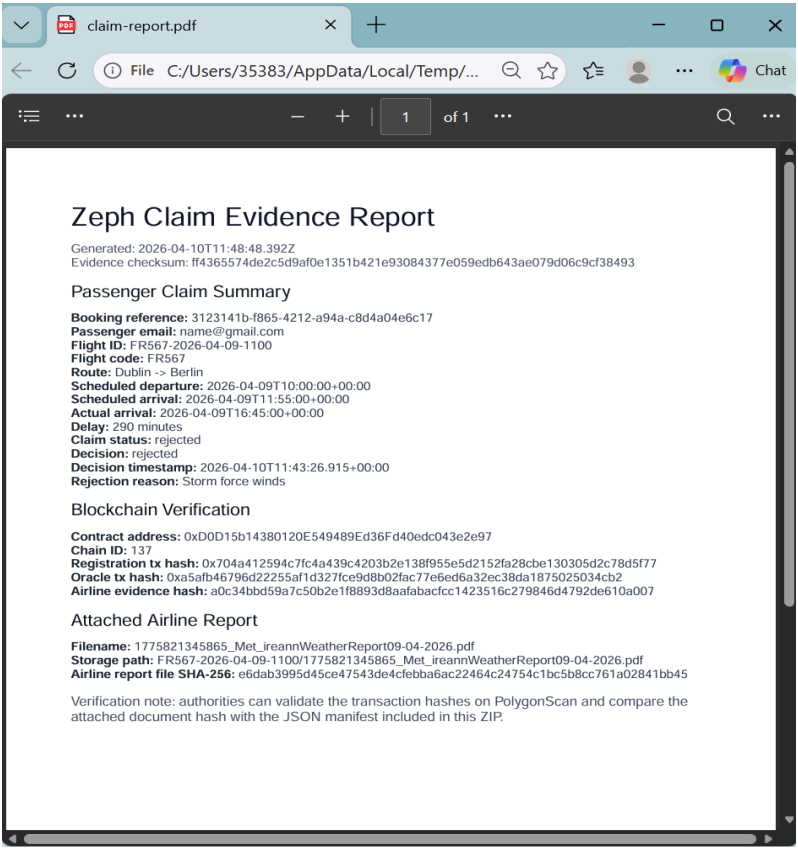


Figure 22 - Claim report PDF

2.8 Claim Acceptance Scenario

If the airline accepts your claim you can verify you have received compensation by navigating to the profile page by clicking the 'Profile' option in the navigation bar

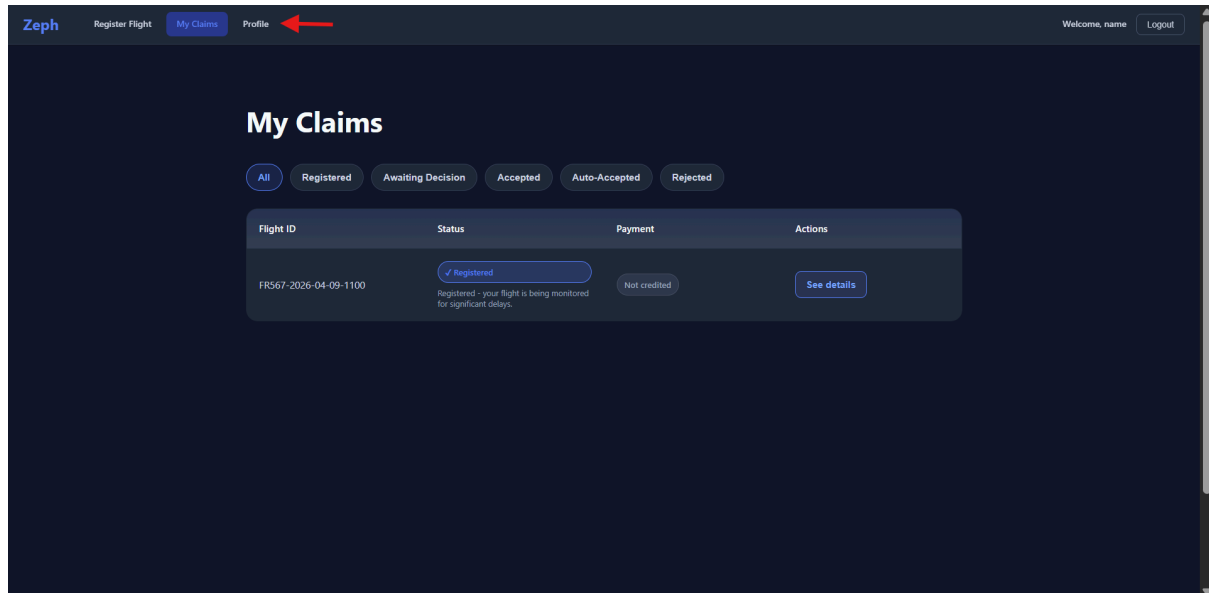


Figure 23 - Click 'Profile' to open profile page

When an airline accepts your claim or if a claim is auto-accepted, your balance will increase by 300 Euro

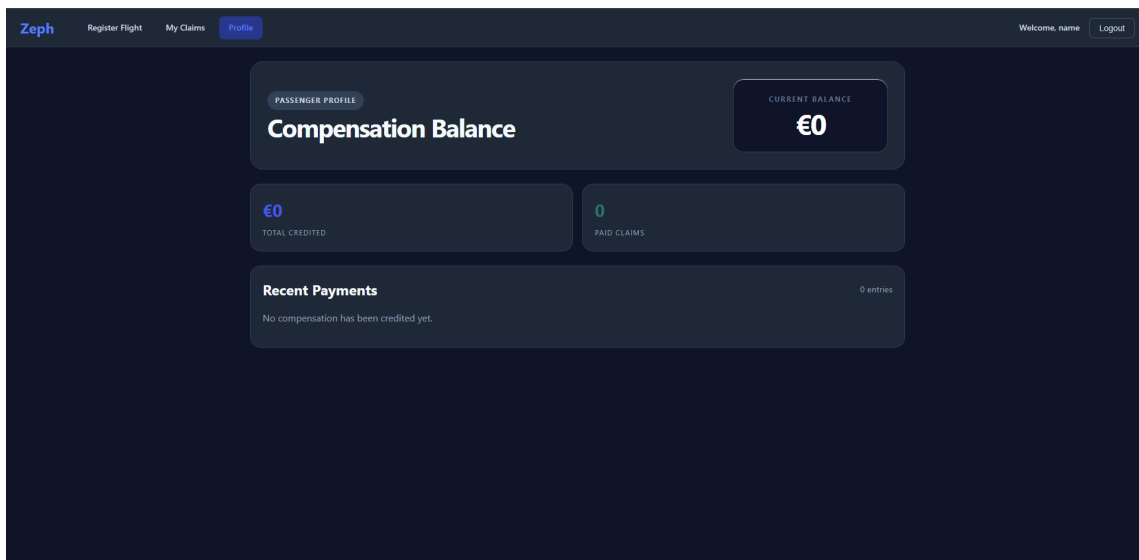


Figure 24 - Profile page with balance displayed

3. Admin Guide

The role of the admin in Zeph is to set the actual arrival times of flights. In practice this information would come from real-world Airport APIs and not an administrator.

Admin staff log in with an email address and password set by Zeph. This prevents an admin from creating an account as a passenger would.

3.1 Logging in

Navigate to the Zeph homepage and click 'Log in'. Enter your admin email address and password, select 'Passenger', and click 'Log In'. You will be directed to the Flight Control dashboard.

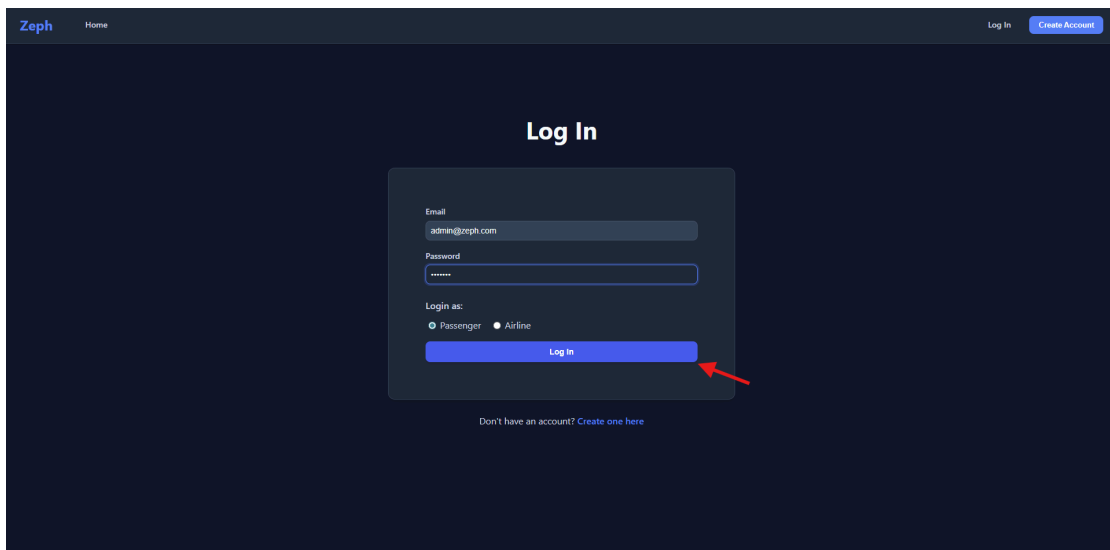
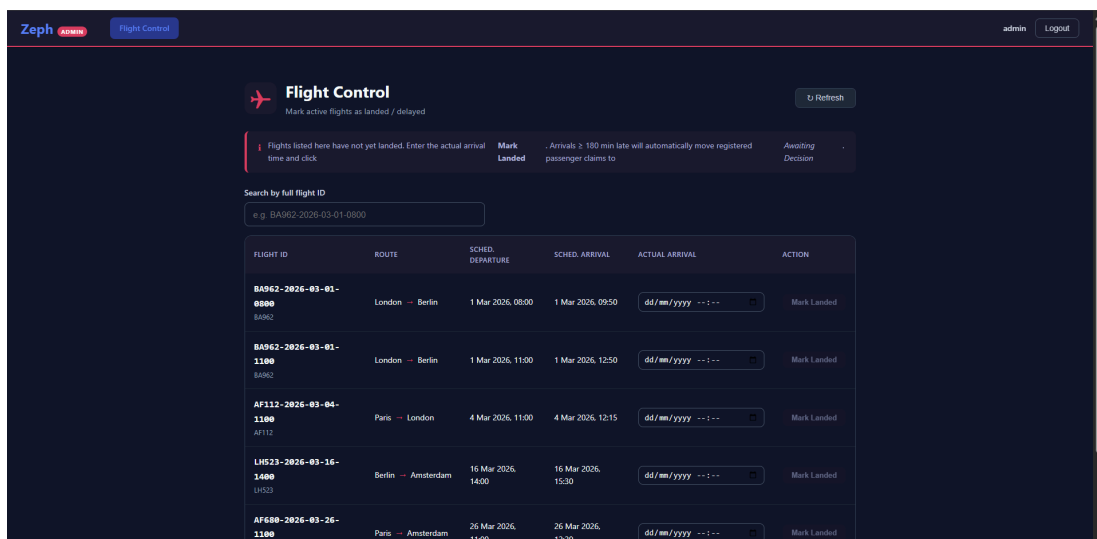


Figure 25 - Login form completed. Click 'Log in'.

3.2 Setting Flight Arrival Times

On the flight control screen search for a specific flight by entering the Flight ID



FLIGHT ID	ROUTE	SCHED. DEPARTURE	SCHED. ARRIVAL	ACTUAL ARRIVAL	ACTION
BA962-2026-03-01-0800 BA962	London → Berlin	1 Mar 2026, 0800	1 Mar 2026, 0950	dd/mm/yyyy --:--	Mark Landed
BA962-2026-03-01-1100 BA962	London → Berlin	1 Mar 2026, 1100	1 Mar 2026, 1250	dd/mm/yyyy --:--	Mark Landed
AF112-2026-03-04-1100 AF112	Paris → London	4 Mar 2026, 1100	4 Mar 2026, 1215	dd/mm/yyyy --:--	Mark Landed
LH523-2026-03-16-1400 LH523	Berlin → Amsterdam	16 Mar 2026, 1400	16 Mar 2026, 1530	dd/mm/yyyy --:--	Mark Landed
AF630-2026-03-26-1100	Paris → Amsterdam	26 Mar 2026, 1100	26 Mar 2026, 1220	dd/mm/yyyy --:--	Mark Landed

Figure 26 - Flight Control dashboard - Enter a flight ID in the search field

Set the 'Actual Arrival' date and time. Click 'Mark Landed' to confirm.

The screenshot shows the Zeph Flight Control interface. At the top, there's a header with 'Zeph ADMIN' and 'Flight Control' tabs. Below the header, there's a 'Flight Control' section with a sub-header 'Mark active flights as landed / delayed'. A message states: 'Flights listed here have not yet landed. Enter the actual arrival time and click Mark Landed. Arrivals 2:180 min late will automatically move registered passenger claims to Awaiting Decision.' Below this, there's a search bar with the text 'FR567-2026-04-09-1100'. A table lists flights with columns: FLIGHT ID, ROUTE, SCHED. DEPARTURE, SCHED. ARRIVAL, ACTUAL ARRIVAL, and ACTION. The first row shows flight FR567-2026-04-09-1100 from Dublin to Berlin, scheduled for 9 Apr 2026, 11:00. The 'ACTUAL ARRIVAL' column is currently empty, and a date picker is open, showing April 2026. The date picker has a grid of days, with the 17th selected. The time '17:45' is also selected. A 'Mark Landed' button is visible next to the flight entry.

Figure 27 - Setting a flight arrival time.

A success banner will appear confirming the arrival time has been set.

The screenshot shows the Zeph Flight Control interface after the arrival time has been set. A green success banner at the top states: '✓ FR567-2026-04-09-1100 marked as delayed by 290 min — claims set to awaiting decision.' Below the banner, the same message as in Figure 27 is present. The search bar still contains 'FR567-2026-04-09-1100'. Below the search bar, there's a message: 'No matching flights. No active flights match the full flight ID 'FR567-2026-04-09-1100'.'

Figure 28 - Flight landing time set, confirmation.

4. Airline Guide

Airline staff log in with an email address on a registered airline domain. This prevents an airline from creating an account as a passenger would. In practice Zeph would create an airline account as needed.

4.1 Logging in

Navigate to the Zeph homepage and click 'Log in'. Enter your airline email address and password, select 'Airline', and click 'Log In'. You will be directed to the Compensation Claims dashboard.

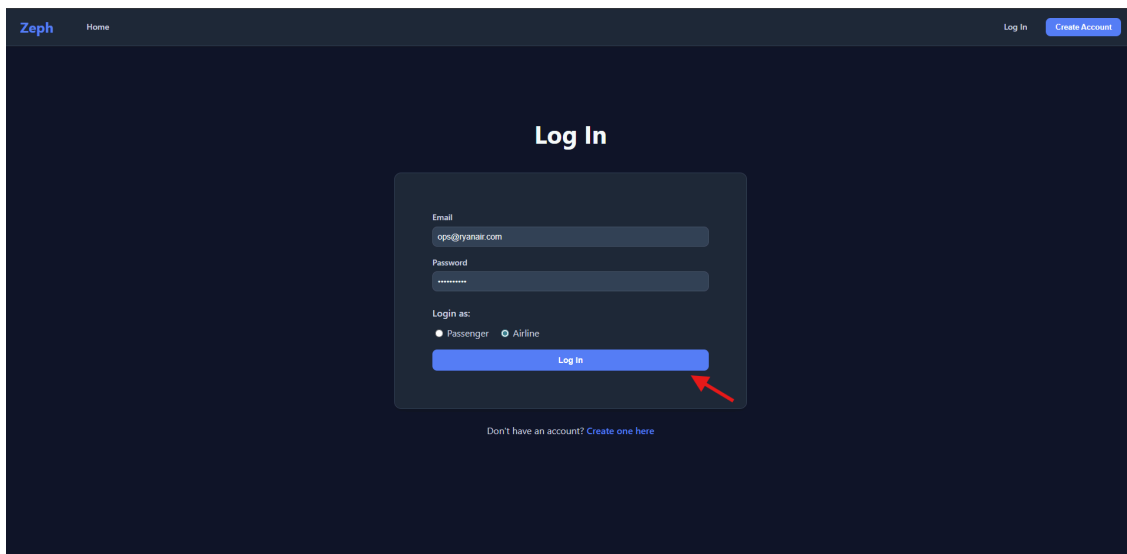
A screenshot of the Zeph login page. The page has a dark blue background. At the top left, the 'Zeph' logo and a 'Home' link are visible. At the top right, there are 'Log In' and 'Create Account' buttons. In the center, there is a 'Log In' heading above a light blue login form. The form contains an 'Email' field with the text 'ops@ryanair.com', a 'Password' field with masked characters, and a 'Login as:' section with radio buttons for 'Passenger' and 'Airline' (which is selected). Below these fields is a blue 'Log in' button, which is pointed to by a red arrow. At the bottom of the form, there is a link that says 'Don't have an account? Create one here'.

Figure 29 - Login form completed. Click 'Log in'.

4.2 Viewing and Managing Claims

The Claims dashboard displays all delayed flights (180+ minutes). Each flight card shows the route, delay duration, scheduled and actual arrival times, and the auto-accept deadline. Displayed flights can be filtered using the buttons at the top of the page. You can filter by; All, Awaiting Decision, Accepted, Rejected.

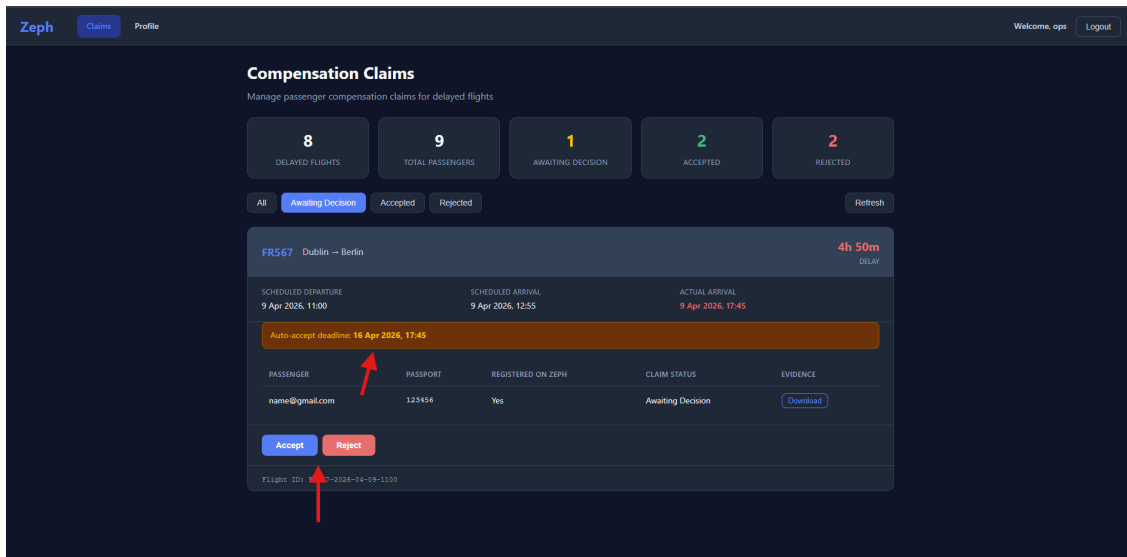


Figure 30- Airline Compensation Claim dashboard.

4.3 Rejecting a Claim

Click 'Reject' on the claim you wish to reject. This will open a claim rejection modal. You must enter the reason for rejection in the 'Evidence Description' field. You must also upload a rejection report PDF in the 'Rejection Report' field. You can also add an optional evidence URL in the 'Evidence URL' field. Then click 'Confirm Reject' to confirm.

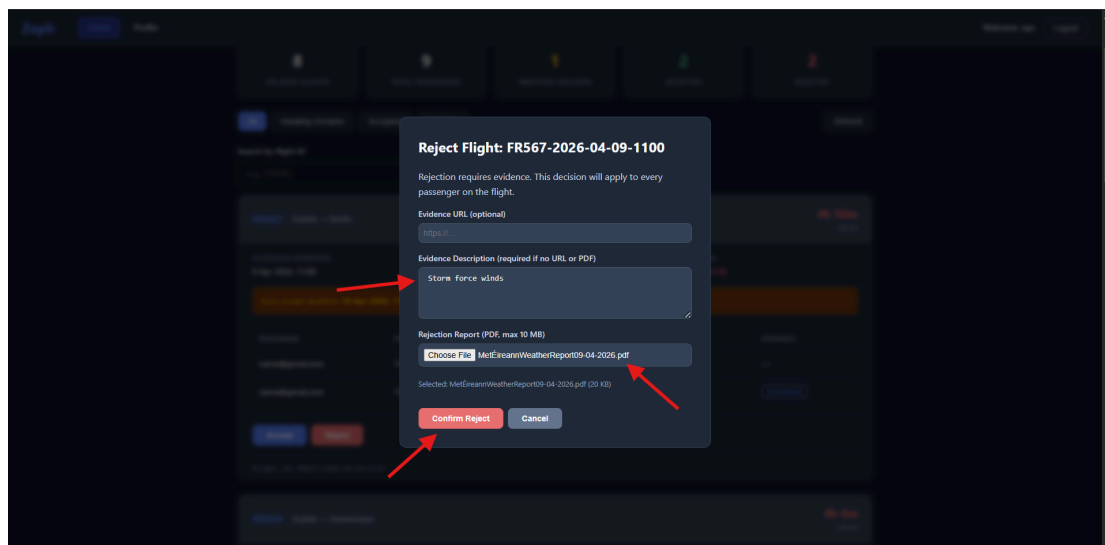


Figure 31 - Claim rejection modal - Enter rejection evidence, click 'Confirm Reject'.

After the rejection is confirmed MetaMask will open and ask you to confirm the blockchain transaction. Review the network and fee details, then click 'Confirm'.

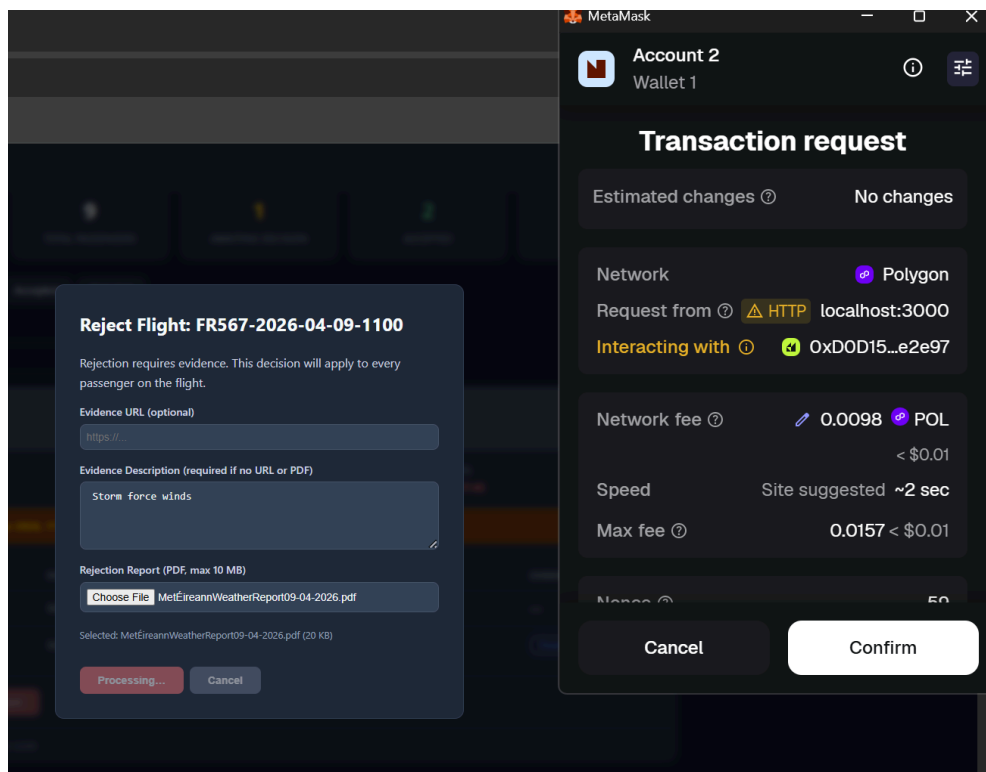


Figure 32 - MetaMask transaction request. Click 'Confirm'

Once the transaction is confirmed on-chain, the page shows a success banner with the transaction hash. MetaMask also confirms the transaction is complete.

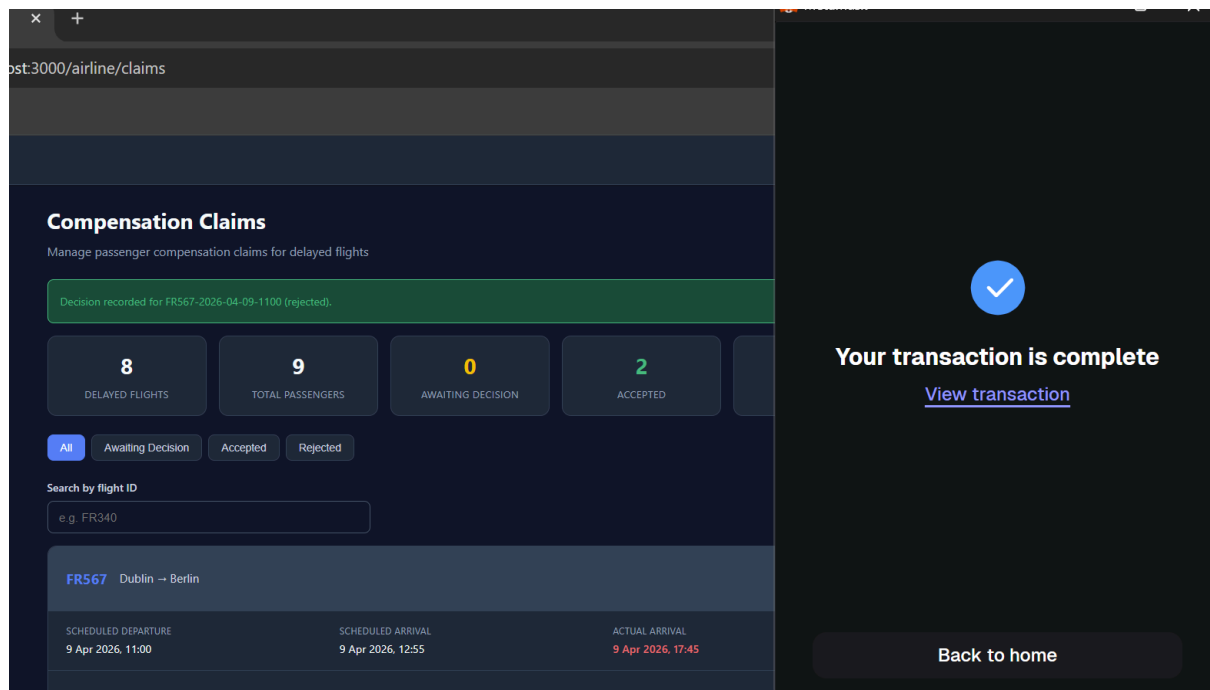


Figure 33 - Transaction confirmed. The Rejection is now registered on the Polygon blockchain.

A claim report can be downloaded by clicking ‘Download’ under ‘Evidence’. The details of this report can be found in section **2.7**.

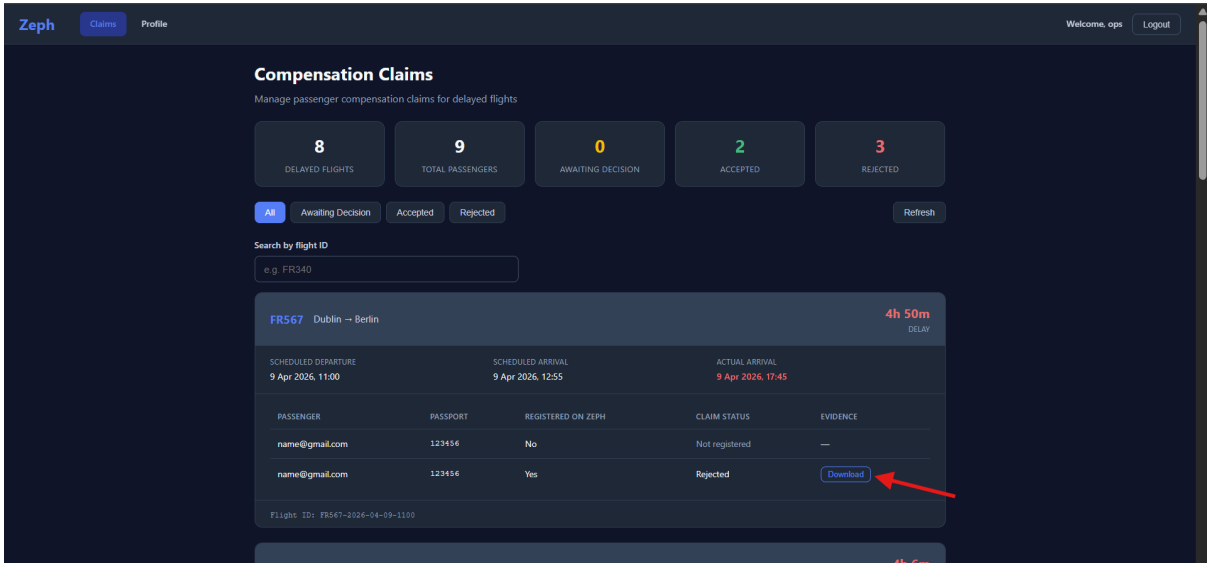


Figure 34 - Click ‘Download’ to download the evidence report ZIP.

4.4 Accepting a Claim

If you wish to accept a claim, simply click “Accept” on the claim. A MetaMask popup will appear as in Figure 32. You must click ‘Confirm’ on this to complete the transaction.

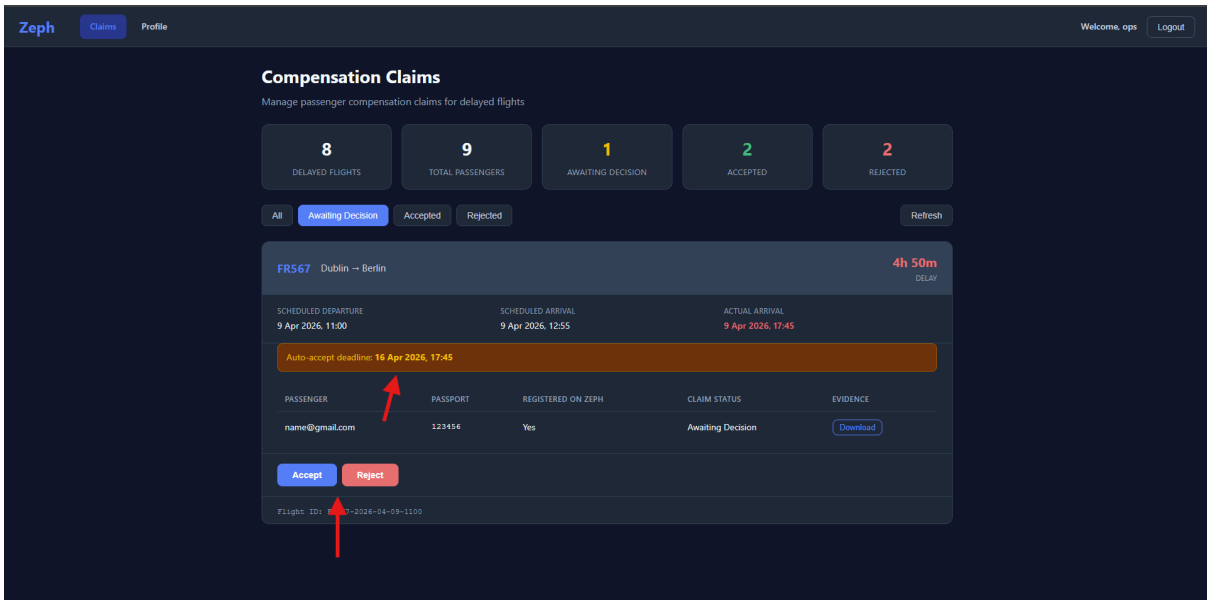


Figure 30- Airline Compensation Claim dashboard. Click ‘Accept’.

Once the acceptance transaction is confirmed on-chain, the affected passengers will receive compensation of 300 Euro. You can keep track of how much your airline has paid in compensation in the ‘Profile’ section. Click on ‘Profile’ in the navigation bar as shown in passenger flow in section 2.8. This page shows the total payout amount as well as counts for; Paid Claims, Accepted, Rejected, Auto-Accepted, Awaiting, and Delayed Flights. As well as a list of recent paid out claims.

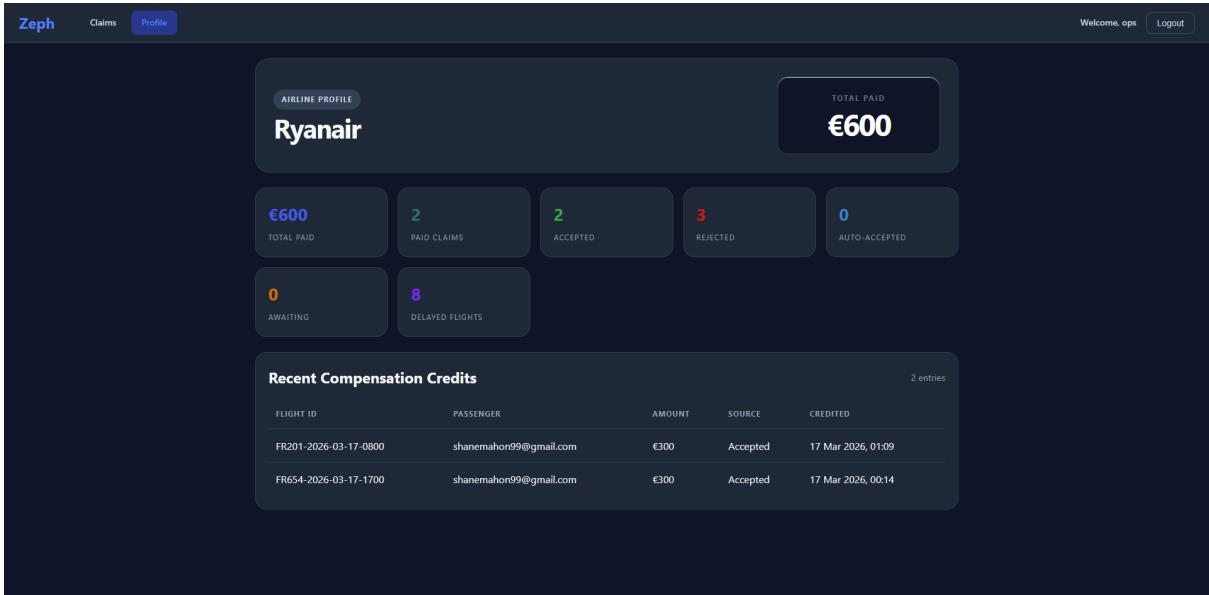


Figure 35 - Airline Profile dashboard.

5. Appendix

5.1 Airline Login Credentials

Airline Email Address Password

Airline	Email Address	Password
Ryanair	ops@ryanair.com	ryanair123
lufthansa	ops@lufthansa.com	lufthansa123
Iberia	ops@iberia.com	iberia123
Easy Jet	ops@easyjet.com	easyjet123
British Airways	ops@britishairways.com	britishairways123
Air France	ops@airfrance.com	airfrance123

5.2 Admin Login Credentials

Email Address	Password
admin@zeph.com	zeph123

5.3 Claim Status Meanings

Registered → Flight is registered on-chain and is being monitored for delays.

Awaiting Decision → Flight was delayed 180+ minutes. Airline has 7 days to respond.

Accepted → Airline has approved the compensation claim.

Auto-Accepted → Airline didn't respond within 7 days, claim automatically accepted.

Rejected → Airline has rejected the claim and provided evidence.